Albert (Iman Mansour)



AI Systems Engineer & Autonomous Agent Architect

> Architecting autonomous agent swarms, sub-second multimodal pipelines, enterprise distributed systems, and physics-driven game engines.

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Military service completed

16+	17	84,000+	4	315K	100%
production systems built	live microservices in UniversalSystem	verified core AI engine LOC	concurrent multimodal threads (WASAPI+GPU)	streamer audience at N5 event	zero-hallucination defensive contract

Executive Summary

applied AI systems & software architecture

I am an **AI Systems Engineer & Distributed Architect** specializing in autonomous multi-agent swarms, sub-second multimodal audio/vision pipelines, and enterprise microservices. My engineering focus addresses the critical gap between raw foundation models and resilient production execution: deterministic state machines, Celery priority queues (High/Default/Bulk), citation-validated vector retrieval (RAG), tamper-proof HMAC audit ledgers, and zero-allocation hot execution loops.

Over the past years, I have architected and shipped **16+ comprehensive software systems** from architectural concept to full production deployment: autonomous agent factories, real-time GPU-accelerated speech-to-speech translation HUDs, adaptive multimodal learning engines, multi-currency accounting ledgers, and physics-driven 3D/2D games.

Core Engineering Pillars

architectural domain expertise

1. Autonomous Multi-Agent Swarms

Orchestrating multi-agent collaboration with Celery priority message queues (0/3/9), human-in-the-loop approvals, and zero-hallucination citation validation against live web sources.

2. Sub-Second Multimodal & Speech

WASAPI loopback audio capture, Butterworth high-pass digital filters, GPU-accelerated Whisper transcribing, Gemini Live bidirectional WebSockets, and OpenCV frame diffing.

3. Enterprise Microservices & Fintech

C# .NET 10 Clean Architecture, FastAPI microservices, double-entry general ledgers, Qdrant vector retrieval, RabbitMQ outbox patterns, and enterprise RTL typesetting engines.

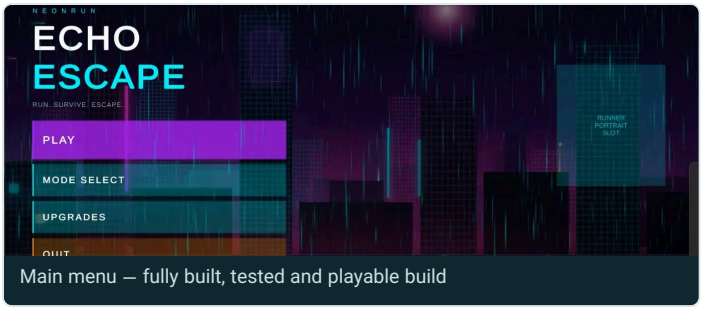
4. Systems Programming & Physics

Electron ASAR binary reverse engineering, SHA-256 base64 re-signing algorithms, strict Zero-GC hot execution loops in game engines, and parabolic projectile trajectory physics.

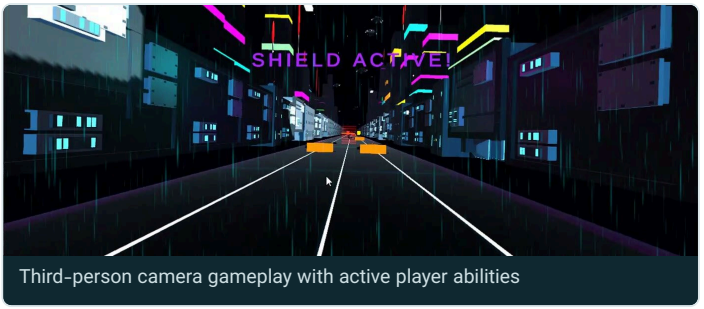
Neon Run: Echo's Escape

3D multi-platform game · flagship gaming project

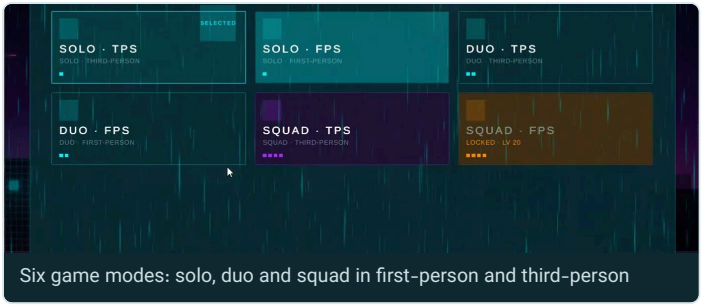
An endless runner set in a cyberpunk future (year 2189): corporations control the city, and a girl named Echo must evade a corporate AI security grid. Built with five levels, six game modes, and two narrative endings. I architected the complete technical core: dual-camera rendering, HUD systems, procedural track pooling, and single-device split-screen multiplayer.



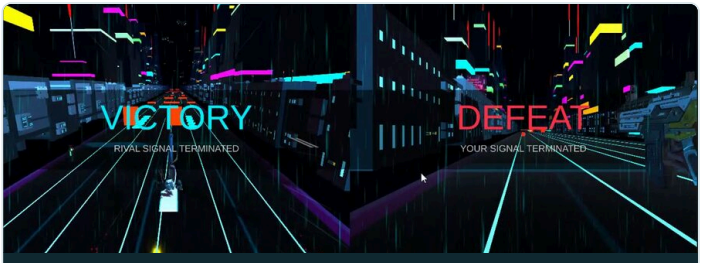
Main menu — fully built, tested and playable build



Third-person camera gameplay with active player abilities



Six game modes: solo, duo and squad in first-person and third-person



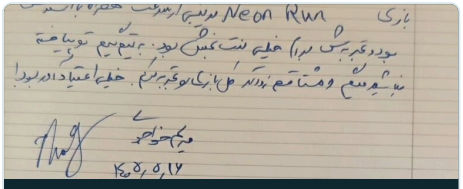
Two-player split-screen mode with dynamic winner resolution

Technical implementation: Dual-camera perspective controllers, live scoring and persistence, special abilities (shield & speed boost), upgrade store, endless procedural track generation with zero heap allocations in Update, single-screen competitive split-screen multiplayer, and identical cross-platform parity on Android, PC, and OpenXR VR.
Status: Preparing for public release with complete narrative progression and bilingual voice lines.

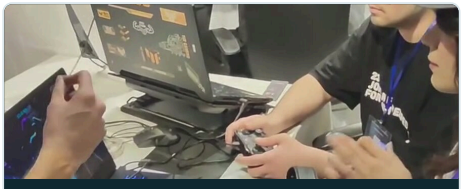
Market Feedback & Public Reception

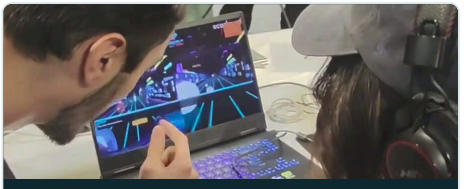
A playable build of Neon Run was showcased on a dedicated booth at the Negative Five accelerator event, where dozens of visitors, streamers, and studio leads playtested the build:

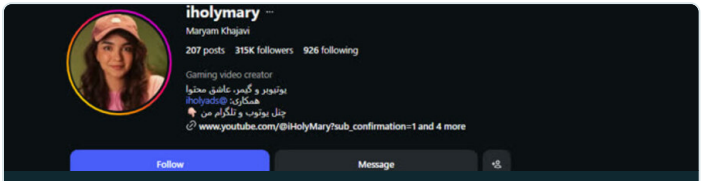
> **"… I really enjoyed the experience … it was seriously addictive!"**
>
> — Maryam Khajavi (iholymary), gaming content creator with over 315,000 Instagram followers and active YouTube channel — after playing Neon Run at the event · handwritten and signed note



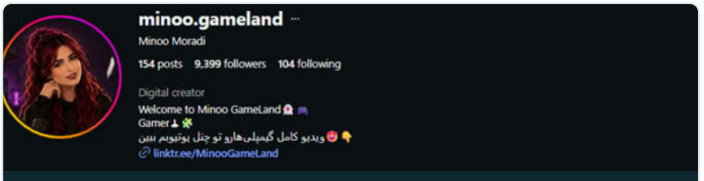
Handwritten, signed note from Maryam Khajavi about the game


Playtesting with gamepad at the booth in front of event media


Continuous uninterrupted gameplay sessions

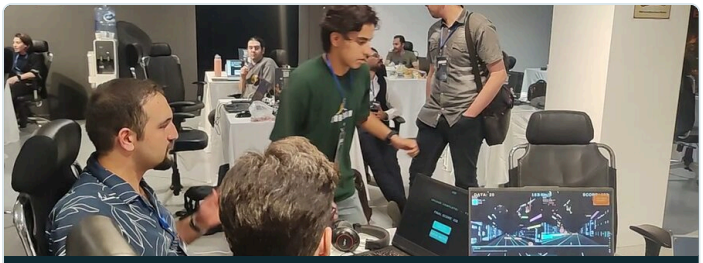


iholymary — 315,000+ followers



minoo.gameland — covered the game via live event story

The director of bazisazbash, one of Iran's veteran game dev communities, playtested the game at the booth and provided positive structural feedback.



The director of bazisazbash playtesting the game at the booth



Gaming content creators covering the game at the event

Maryam Khajavi — 315K

Instagram profile of the streamer who played the game

Minoo Gameland

Streamer story featuring the game from Negative Five

Bazisazbash

The studio whose director verified the game

NitroLango Live Build

Interactive language-learning web app — online now

UniversalSystem — Autonomous AI Software Factory

distributed AI systems

UniversalSystem is an enterprise multi-agent orchestration platform that automates software generation, multimodal synthesis, automated test suites, and long-term project memory. Architected with isolated containerized microservices, priority Celery message queues, cryptographically tamper-proof HMAC ledgers, and automated failover bridges between cloud and local model sessions. The live health verification is shown on the next page.

84,290	17 / 17	385+
lines of code across 432 files	live containerized services healthy	automated test suites passing

Architectural Components

role of each microservice

api-gateway YARP reverse proxy: centralized auth, rate limiting, and intelligent load balancing.

dotnet-api C# .NET 10 Clean Architecture: core domain models, MassTransit outbox, banking rules.

celery-worker Distributed background workers with message priority tiers: **High (0)**, **Default (3)**, **Bulk (9)**.

sqlserver Relational transactional store: structured schemas, financial audits, and enterprise state.

rabbitmq Enterprise message broker: reliable outbox delivery ensuring zero message drops.

prometheus Metrics collector: gathers performance metrics across all microservices continuously.

jaeger Distributed tracing: end-to-end request lifecycle analysis from gateway to database layers.

web-frontend Next.js 15 enterprise console: interactive UI for steering autonomous agents.

python-ai AI processing core: multi-agent orchestration, RAG retrieval, document pipelines.

node-microservices Event-driven services: real-time device communication and MQTT protocol bridge.

qdrant Vector database: dense embeddings and cosine similarity search with exact match thresholds.

redis In-memory cache with AOF persistence: agent state caching, session management, rate limits.

host-bridge Local Windows bridge: routes overflow requests to authenticated Claude/Codex/Antigravity CLIs.

grafana Observability dashboards: real-time tracing of throughput, token latency, and error rates.

vision-watcher AI vision sentinel: inspects graphical and UI outputs to flag rendering regressions.



Empirical proof: live system health check with all 17 containerized services verified healthy with passing probes

Key Architectural Capabilities & Defensive Design

- **Cryptographic HMAC Audit Ledger:** Every agent execution step and administrative decision is logged into an immutable, HMAC-chained hash ledger preventing tampering.
- **Multi-Model Cascade & Local Host Bridge:** Automated fallback when cloud APIs encounter rate limits (429/503), gracefully shifting traffic to local authenticated developer CLI sessions.
- **PII Redaction & SSRF Guardrails:** Automated regex-based scrubbing of credit cards, national IDs, and IBANs, paired with strict outbound connector egress filters against intranet IP ranges.
- **Universal Doc Factory:** 5-stage sequential agent pipeline (Brief ➔ Research ➔ Architecture ➔ Writer ➔ Reviewer) featuring zero-hallucination citation validation against live web sources and Persian/RTL typesetting export to PDF, DOCX, and PPTX.
- **Microsoft AI Architect Academy:** 8-level AI system architecture curriculum featuring advanced Semantic Kernel patterns, AutoGen swarm swarms, GraphRAG knowledge graphs, and MCP server tools in C# .NET.
- **Headless Editor Agent Interop:** Dynamic-port Model Context Protocol (MCP) bridges enabling autonomous agents to inspect, manipulate, and compile game projects headlessly.

Sub-Second Real-Time Multimodal & Speech Engines

RapScribe-AI (OmniScribe HUD)

Low-latency transcription & neural translation: Windows WASAPI loopback audio capture ➔ Butterworth HPF & RMS gating ➔ GPU-accelerated faster-whisper large-v3-turbo ➔ Gemini 3.8 Flash cascade across 4 concurrent non-blocking worker threads.

GeminiLiveCoPilot

Frameless always-on-top desktop HUD over Gemini Live bidirectional WebSocket API: 1fps desktop capture with MSE and SSIM frame-diff deduplication to minimize token consumption, with 16kHz PCM input and 24kHz synthesized output.

NitroLearnAI (NitroLango)

Adaptive language learning platform covering the complete Menschen CEFR corpus (72 Lektionen from A1.1 to B1.2), 77 interactive German grammar engines, SuperMemo SM-2 spaced repetition decay formula, and resilient browser speech watchdogs.

German-Neural-Immersion

Automated multimodal course studio parsing textbook PDFs and audio archives into interactive sentence builders (Satzbau) and dual-language synchronized audio lessons synthesized via ElevenLabs neural voices.

Enterprise SaaS, Fintech, Open Source & Commercial Projects

ZarDigit — Enterprise Financial & Accounting Suite

Role: Systems Architect & Lead Full-Stack Engineer | Next.js 15 · TypeScript · FastAPI · PostgreSQL | Private Enterprise Platform

- Architected a multi-currency double-entry general ledger with high-precision decimal arithmetic, trial balances, and transaction audit trails.
- Implemented automated financial statement generation (Balance Sheets, Profit & Loss, Cash Flow) with native RTL typesetting support.

Zoho Books Clone — Enterprise Double-Entry Cloud Accounting Suite

Role: Systems Architect & Full-Stack Engineer | Next.js 15 · React 19 · TypeScript · Material-UI · Multi-Tenancy | Private Enterprise Platform

- Full-featured platform mirroring international Zoho Books standards: 118 static pages compiled with zero errors under Next.js 15 App Router.
- Engineered a zero-leakage 20-language internationalization (i18n) engine across 436 native keys, full double-entry accounting, bank feeds & RBAC.

AuroSocialEngine — Sovereign Decentralized Social Engine

Role: Data Architect & Distributed Systems Engineer | Python · FastAPI · Node.js · Redis · WebSockets | Private Enterprise Repository

- Privacy-first decentralized social engine featuring sovereign messaging, cryptographic identity verification based on public/private key pairs.
- Engineered feed curation scoring algorithms with automated bot detection, anti-spam filters, and high-throughput distributed event streaming.

antigravity-scroll-unpin — Reverse Engineering & Open Source Tool ⭐

Role: Tool Architect & Author | Node.js · PowerShell · Electron ASAR · SHA-256 Digest | Public Open Source Repository

- Deconstructed Electron binary app.asar archive in Google Antigravity to unpin and disable sticky headers within chat panels.
- Implemented the exact VS Code product.json cryptographic SHA-256 base64 digest calculation algorithm, eliminating corrupt installation warnings.

Arash Kamangir — Commercial Mobile Game

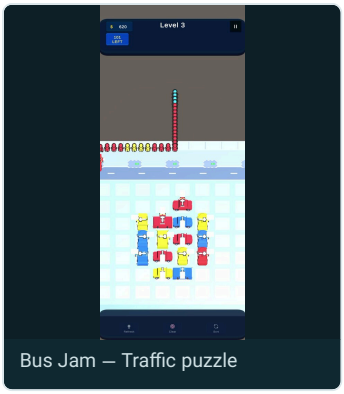
Role: Lead Gameplay & Tools Programmer | Ferdowsi University Science & Technology Park | 2024 – present

- Implemented parabolic arrow trajectory physics, dynamic bow tension mechanics, and multi-joint stickman ragdoll collisions (Box2D).
- Developed custom Unity Editor tools for visual wave layout; optimized physics loop for locked 60 FPS on budget devices.

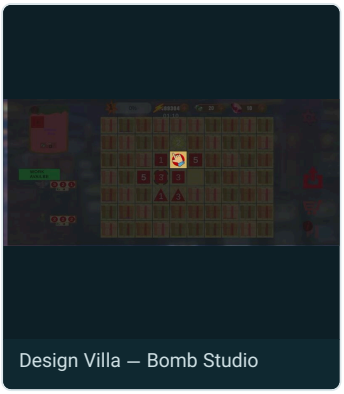
Bus Jam Case, TravelTown, Parsa & Design Villa — Commercial Titles & Engines 🚀

Role: Algorithm Designer & Systems Developer | Bomb Game Studio & Independent | Published on Cafe Bazaar & Myket

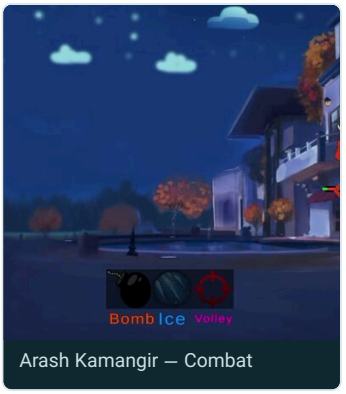
- **Parsa (Cafe Bazaar) & Design Villa:** Action & decor games for Bomb Studio; gameplay logic, cloud saves & FSM.
- **Bus Jam Case & TravelTown:** Multi-agent A* grid pathfinding for vehicle queues, merge generators & binary state.



Bus Jam — Traffic puzzle


Design Villa — Bomb Studio


Parsa — Commercial title



Arash Kamangir — Combat

National Honors, Industry Recognition & Elite Talent Credentials

Negative Five International Accelerator — Third Cycle National Competition · 2026

National Award Winner

Top National Ranking & Award Winner in the accelerator entry competition showcasing Neon Run: Echo's Escape; awarded official certificate and honor on stage signed by the Accelerator CEO and Xelerate VR Representative

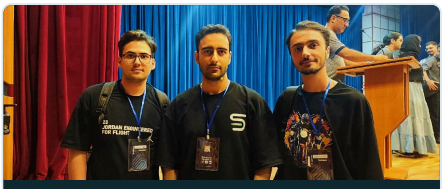
- **National Award & Official Recognition:** Won top national honors among competing engineering and interactive systems; formally honored on stage during the closing ceremony.
- **Elite Talent & Global Talent Visa Evidence:** Possesses substantial legal and evidentiary value for National Elites evaluation and international Global Talent / Exceptional Talent immigration streams (UK Tech Nation, US O-1A / EB-1A, EU Talent) as primary proof of recognized national achievement.
- **Dedicated Live Showcase Booth:** Operated a fully equipped live playtest booth throughout the event; audited and playtested on controller by industry leaders, judges, and studio heads.
- **Mass Audience & Media Coverage:** Live broadcast and featured coverage by prominent gaming creators reaching 315,000+ active followers, accompanied by signed handwritten testimonials.



Dedicated national competition booth

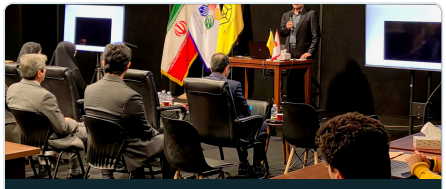


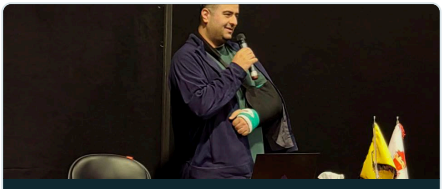
Official stage certification & award


Main conference venue & engineering team

GameDev Weekend / Game Dojo Summit — November 2025

Participated in the two-day intensive developer convention at Mashhad Innovation Factory


Main conference auditorium


Technical keynotes and discussions


Industry challenges and engineering panel

Mentorship, Knowledge Transfer & Video Archive

in-person & online archive

Role: Private Systems & Game Programming Instructor | Unity Engine & C# / Python | In-person sessions, Mashhad

Mentorship centered on commercial software architecture: memory layout fundamentals, component-based design patterns, connecting AI models via the Model Context Protocol (MCP), zero-allocation performance optimization, and final compilation for Android and Windows targets.

Video Archive

19 videos · 56 minutes · playable online in any browser

High-resolution recorded footage exists for every major system and game documented in this resume, hosted online without requiring any local installations:

Neon Run: Echo's Escape — 6 videos

Solo first-person · solo third-person · two-player split-screen · full 7-minute uncut run · special abilities · endless procedural generation

Commercial games — 2 videos

Arash Kamangir · Bus Escape

Negative Five event — 9 videos

Maryam Khajavi (315K) playtesting at the booth · studio director walkthrough · dedicated booth tour · stage certificate award · venue atmosphere

NitroLango — 2 videos

Comprehensive platform walkthrough · latest update



Online Video Archive

albert392392.github.io/portfolio



Live Product Demo

Open the NitroLango web app in your browser right now

The 5-Point Empirical Verification Gate

defensive architecture manifesto

Core Engineering Philosophy: A system reporting success for work it did not empirically verify is a failure. Every system I build is governed by five mandatory defensive rules:

- **1. Empirical Proof Mandate:** Zero speculation. No task is certified without exit code 0, 0 compilation errors, exact file paths and line ranges, and clean live verification.
- **2. Defensive State Bounds:** Mandatory validation guards checking for NaN and Infinity prior to mutating spatial transforms, physics vectors, or financial balances.
- **3. Hot-Loop GC Prevention (Zero-GC):** Zero object instantiations (new), zero string concatenations or LINQ queries, and pre-cached Property IDs in per-frame update loops.
- **4. Resilient Watchdogs:** Universal watchdog timeouts and stall recovery mechanisms guarding WebSockets, browser speech APIs, and asynchronous message queues.
- **5. Fail-Closed Security:** Length-only secret inspection, strict credential isolation, SSRF-guarded outbound sinks, and cryptographically chained HMAC audit logs.

Education, Formal Certifications & Engineering Bootcamps

Technical & Vocational High School Diploma — Computer Network & Software — June 2024

Shahed Isar Vocational High School, Mashhad — In-depth systems architecture, OS internals, networking & databases

Advanced C++ Systems Engineering, Algorithms & Data Structures Bootcamp (Faravin) — 2021

Vocational High School Internship (11th Grade) — Algorithmic design, number systems, low-level memory allocation & complexity analysis

Web Development & Database Systems Engineering — Official State Certificate — 450 Hours

National Technical & Vocational Training Organization — Internationally accredited standard; web architectures, APIs & data pipelines

Game Dojo Specialized Interactive Systems & Game Engineering Bootcamp — Comprehensive

Real-time architecture with Unity & C#, event-driven modular design, Zero-GC allocation patterns & profiling

Advanced Backend Architecture with ASP.NET Core & C# .NET — 400 Hours

Clean Architecture, event-driven microservices, MassTransit/RabbitMQ distributed messaging & Transactional Outbox

Relational Database Architecture & Administration with SQL Server — 360 Hours

Advanced indexing strategies, execution plan tuning, ACID transactional integrity & concurrency management

Computer Networking & Infrastructure Engineering — Network+ — 450 Hours

Enterprise network topologies, TCP/IP stack mechanics, low-level socket programming & distributed client/server security

Comprehensive Professional English Language Programs (Childhood to Present) — Multi-Term

Shokouh Language Institute — Consecutive honors transcripts (scores 91–100); fluent technical writing & international communication

Technical Competency Matrix

Python (FastAPI, AsyncIO, Celery) | C# (.NET 10 Clean Architecture) | TypeScript / Next.js 15 | Multi-Agent Swarms | Semantic Kernel | AutoGen | Model Context Protocol (MCP) | Qdrant Vector DB | RAG Architecture | faster-whisper (CUDA) | Gemini Multimodal Live API | WASAPI Audio Loopback | OpenCV Vision Diffing | Electron ASAR Patching | Docker Compose | RabbitMQ / MassTransit | Redis AOF | SQL Server / PostgreSQL | Prometheus / Grafana / Jaeger | Unity 6.3 (URP, Zero-GC) | Box2D Ragdoll Physics | Git Plumbing

Languages

Persian: Native — professional technical writing, architecture documentation, and system specifications

English: Professional working proficiency — deep architectural whitepapers, codebases, and global communication

German: Intermediate (Active B1 learner) — solid grasp of Menschen CEFR grammar structures and multimodal audio synthesis

Albert (Iman Mansour) — AI Systems Engineer & Autonomous Agent Architect

Open for high-impact AI systems engineering, distributed enterprise architectures, and cutting-edge software roles

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